DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

B.S. Abdur Rahman Crescent Institute of Science & Technology
Deemed to be University u/s 3 of the UGC Act, 1956

EcoSort Intelligence: Smart E-Waste Classification & Economic Optimizer

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Domain of project : Artificial Intelligence

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EcoSort Intelligence: Smart E-Waste Classification & Economic Optimizer

Description :

- EcoSort Intelligence is an innovative framework that leverages deep learning (ResNet, EfficientNet) for automated, accurate classification of e-waste components and market basket analysis (MBA) to predict the economic viability of recycling versus reuse.
- By optimizing resource recovery and prioritizing high-yield component clusters, it enhances sustainability, reduces environmental impact, and maximizes profitability in e-waste management.

Abstract

- **Automated Component Classification:** Utilizes deep learning models (ResNet, EfficientNet) to accurately identify and count laptop parts, streamlining e-waste sorting.
- **Market Basket Analysis (MBA):** Analyzes historical data to uncover component co-occurrence patterns and predict recycling vs. reuse outcomes.
- **Economic Descriptive Modeling:** Evaluates costs, market values, and conditions to determine the most profitable processing strategy for e-waste.
- **Sustainability & Circular Economy:** Promotes resource recovery and reduces environmental impact by optimizing high-yield component clusters and advancing sustainable e-waste practices.

Existing and proposed work

Aspect	Existing Work	Proposed Work
AI Techniques	Various models like CNNs, AlexNet, Inception, and hybrid transfer learning	Deep learning with ResNet and EfficientNet for precise classification
Prediction Focus	E-waste classification and sorting, waste generation prediction	Market Basket Analysis (MBA) for economic feasibility of recycling vs. reuse
Accuracy	Models achieve accuracy between 83% to 99.1%	Combines deep learning and MBA for optimal component clustering and decision-making
Sustainability	Emphasizes automation for efficiency and AI-driven classification	Optimizes e-waste processing by prioritizing high-yield clusters, promoting cost-effectiveness
Circular Economy	Uses AI for better e-waste identification and sorting	Advances circular economy with integrated component classification and predictive analytics

IPO

Input:

- Images of components
- Component Details:

Type: (CPU, GPU, RAM, PSU, SSD, etc.)

Brand & Model: (e.g., Intel i7-8700, RTX 3060

Condition: (New, Used, Damaged, Repairable)

Age: (Number of years since purchase)

```python
import torch
import torch.nn

class dataloader():

    def __init__(self,train_dir,test_dir,train_size,test_size,val_size,batch_size,seed = 42):
        self.test_dir = test_dir
        self.train_dir = train_dir
        self.train_size = train_size
        self.test_size = test_size
        self.batch_size = batch_size
        self.val_size = val_size
        self.seed = seed

    def load_data(self):
        print("loading data.....")
        training_set = datasets.ImageFolder(self.train_dir, transform=self.transform)
        trainsize = int(round((1 - self.val_split) * len(training_set)))
        valset_size = len(training_set) - trainsize

        trainset, valset = random_split(training_set, [trainsize, valset_size], generator=torch.Generator().manual_seed(self.seed))
        testset = datasets.ImageFolder(self.test_dir, transform=self.transform)

        trainloader = DataLoader(trainset, batch_size=self.batch_size, shuffle=True)
        valloader = DataLoader(valset, batch_size=self.batch_size, shuffle=True)
        testloader = DataLoader(testset, batch_size=self.batch_size, shuffle=True)

        return trainloader, valloader, testloader
```

IPO

Process: Algorithms Service - 1

1. Preprocessor (preprocessor.py)
- Image preprocessing (denoising, contrast adjustment)
- Extract features if needed
- Data augmentation (rotation, flipping, cropping)

2. Dataloader (dataloader.py)
- Load images & labels (classification + count)
- Apply transformations (resize, normalize, augment)
- Create PyTorch Dataset & DataLoader

3. Model (model.py)
- Load pre-trained **ResNet** / **EfficientNet** models
- Modify output layer for classification & counting
- Define loss function & optimizer

4. Main (main.py)
- Load dataset & initialize model
- Train, validate, and test
- Save best model & generate reports

```
Project Root
|— src
|    |— dataloader.py
|    |— preprocessor.py
|    |— model.py
|    |— main.py
|— data
|— models
|— results
```

IPO

Process: Algorithms Service- 2 (Profit analysis)

1. **User Input** → Component details (Type, Brand, Condition, Age).
2. **Market Data Collection** → Fetch resale prices, metal values, repair & recycling costs.
3. **Profit Calculation** →
   Reuse Profit = Resale Price - (Repair + Logistics).
   Recycle Profit = Extracted Material Value - Processing Cost.
4. **Decision Making** → Suggest **Resale or Recycling** based on higher profit.
5. **Future Price Prediction** → Use **Machine Learning (ARIMA, LSTM)** for trends.
6. **Final Output** → Display best profitability option & suggested action.
7. **Check for updated prices** → Using scrapers and api find the current market price and update the model.

IPO

Process: Algorithms Service - 2 (MBA)

1. **Data Collection** → Analyze past purchases & frequently bought items.
2. **Pattern Mining** → Use **Apriori / FP-Growth** to find best-selling combos.
3. **Confidence & Lift Scores** → Rank purchase trends.
4. **Recommendation Generation** → Suggest **best PC component bundles** for buyers.
5. **Final Output** → Show **"Frequently Bought Together“** in a table format.

IPO

Output: Performance evaluation measure

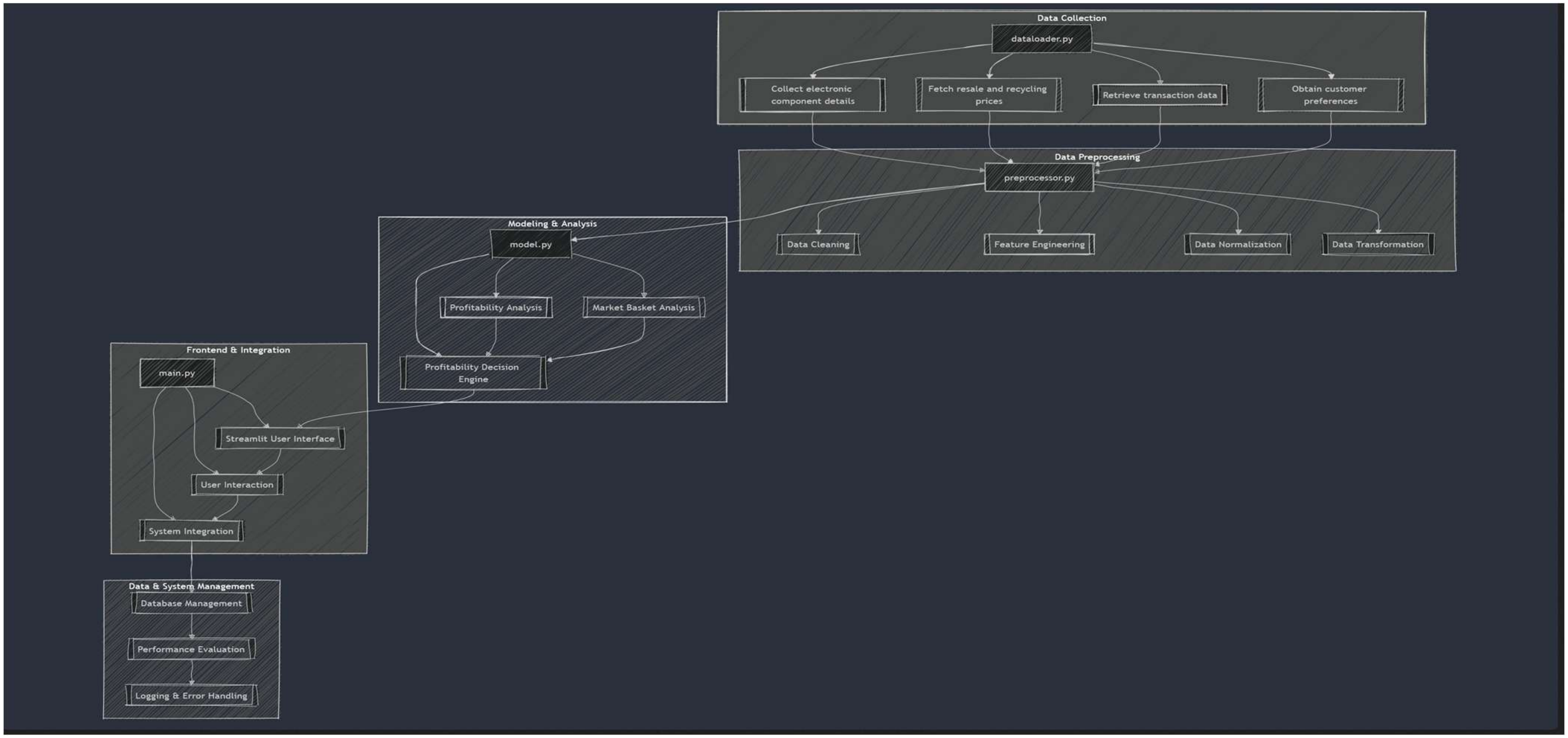
Metric	Description	Purpose
Profit Prediction Accuracy	Measures how accurately the system predicts whether recycling or reusing is more profitable.	Evaluate the reliability of profitability predictions.
Precision (for Reuse/Recycle/Image classification)	Measures the proportion of correctly predicted profitable reuse or recycle decisions.	Assess the accuracy of reuse vs. recycle decision-making.
Lift (Market Basket Analysis)	Measures how much more likely two components are bought together than expected by chance.	Evaluate the strength of association between items.
AUC (Area Under ROC Curve)	Measures the area under the Receiver Operating Characteristic curve.	Evaluate the trade-off between true positives and false positives in classification.

IPO

Output: Performance evaluation measure

Metric	Description	Purpose
Processing Time (System Efficiency)	Time taken to process input data and generate results for profitability and market analysis.	Assess system performance and user experience.
Accuracy	Percentage of total correct predictions to wrong predictions	To assess the models Performance in classifying the images
F1 score	Harmonic mean of precision and recall to provide a balanced measure.	Evaluate the balance between precision and recall.

Architecture Diagram



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Modules

```
├── 1 dataloader.py
│   ├── Load Component Data
│   ├── Fetch Market Prices
│   ├── Retrieve Transaction History
│   └── Collect Customer Preferences
├── 2 preprocessor.py
│   ├── Data Cleaning
│   ├── Feature Engineering
│   ├── Data Normalization
│   ├── Data Transformation
│   └── Image Preprocessing
│       ├── Resize Images
│       ├── Normalize Pixel Values
│       └── Data Augmentation (if necessary)
├── 3 model.py
│   ├── Profitability Analysis
│   ├── Market Basket Analysis
│   ├── Profitability Decision Engine
│   └── Image Classification & Part Counting (ResNet + EfficientNet)
│       ├── ResNet (Image Classification)
│       │   ├── Classify Components (e.g., CPU, GPU)
│       │   └── Fine-Tune ResNet for Component Recognition
│       ├── EfficientNet (Part Counting)
│       │   └── Count Components in Image (e.g., number of parts)
│       └── Combined Model
│           ├── Integrate ResNet and EfficientNet for Classification + Counting
│           └── Return Class and Count Results
│
```

Modules

```
├── main.py
│   ├── Streamlit User Interface
│   ├── User Interaction
│   ├── Visualization
│   └── System Integration
└── Data & System Management
    ├── Database Management
    ├── Performance Evaluation
    └── Logging & Error Handling
```

Software

1. Programming Language:
   Python, , go (Optional)

2. Data Handling & Analysis:
   Pandas, NumPy
   Scikit-learn (for market basket analysis)

3. Deep Learning Libraries:
   PyTorch (for ResNet, EfficientNet)

4. Image Processing:
   OpenCV, Pillow

5. Market Basket Analysis:
   mlxtend (for Apriori algorithm)

6. Web Interface:
   Streamlit/HTMX

7. Database/Storage:
   SQLite / MySQL / PostgreSQL / MongoDB

8. Version Control:
   Git (GitHub, GitLab, or Bitbucket)

Front End Design

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